

## **JAVA INTERVIEW QUESTIONS**

### **1. What is Java?**

Java is an object oriented programming language and high-level programming language which is used to develop software.

### **2. List features of Java?**

Some features include Object Oriented, Platform Independent, Robust, Interpreted, Multi-threaded.

### **3. What are the supported platforms by Java Programming Language?**

Java runs on a variety of platforms, such as Windows, Mac OS, and the various versions of UNIX/Linux like HP-Unix, Sun Solaris, Redhat Linux, Ubuntu, CentOS, etc.

### **4. What is JVM ? Why is Java called the Platform Independent.**

It is whole responsible to execute the java program. JRE is installed in each and every electronics device and .class file can be executed on any electronic device , hence Java is platform independent.

### **5. What is the Difference between JIT , JDK and JRE ?**

It is whole responsible to convert . class file to binary format.

It is whole responsible to execute the java program.

It is an environment setup provided to run the java program.

### **6. List two Java IDE's?**

NetBeans, Eclipse, etc.

### **7. List some Java keywords unlike C,C++ keywords?**

Some Java keywords are import, super, finally, etc.

### **8. What are the Types by Java ?**

1) Class    2) enum    3) Annotation    4) Interface

**9. What are the Data Types supported by Java ?**

byte • short • int • long • float • double • Boolean • char

**10. What is the default value of byte datatype in Java?**

Default value of byte datatype is 0.

**11. What is the default value of float and double datatype in Java?**

Default value of float and double datatype is float its 0.0f ( unique ) and for double it's 0.0d

**12. When a byte datatype is used?**

This data type is used to save space in large arrays, mainly in place of integers, since a byte is four times smaller than an int.

**13. What is Variable?**

Variable is a named memory location which is used to store some values or data and it can change N no of times during execution.

**14. What kind of variables a class can consist of?**

A class consist of Local variable and global variables.

**15. What is a Local Variable?**

Any variables which is declared within the method is called as local variables.

**16. What is a global Variable?**

Any variables which is declared outside method and inside the class is called as Global variables.

**17. What is Method?**

Method is a block of statements which will get executed whenever it is called or invoked.

**18. What is Static ?**

Any member of the class declared with a keyword static is called as static member of the class.

**19. What is Non - Static ?**

Any member of the class declared without a keyword static is called as Non - static member of the class.

**20. Define composition?**

The class having an object of another class is known as composition. It is also called HAS a relationship.

**21. What is static initialization block?**

Any block which is declared with a keyword static is called as static initialization block . SIB is used to initialize static members. SIB will get executed before main method.

**22. What is Non - static initialization block?**

Any block which is declared without a keyword static is called as Instance initialization block. IIB is used to initialize non - static members. IIB will get executed whenever an object is created.

**23. What are pass by value and pass by reference ?**

Calling or invoking a method by passing primitive type of data is called as call by value or pass by value.

Calling or invoking a method by passing references variables is called as call by references or pass by references.

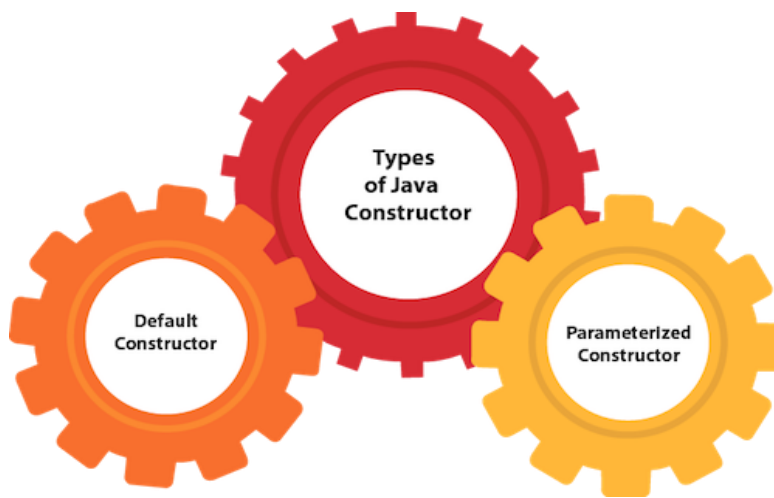
**24. What is a Constructor ?**

It is a special type of method or special type (member) of the class which is used to initialize data members.

**25. How many types of constructors are used in Java?**

There are two types of constructors in Java.

- Default Constructor: default constructor is the one which does not accept any value.
- The default constructor is mainly used to initialize the instance variable with the default values. It can also be used for performing some useful task on object creation. A default constructor is invoked implicitly by the compiler if there is no constructor defined in the class.
- Parameterized Constructor: The parameterized constructor is the one which can initialize the instance variables with the given values. In other words, we can say that the constructors which can accept the arguments are called parameterized constructors.



## **26. What is the purpose of default constructor?**

The java compiler creates a default constructor only if there is no constructor in the class.

## **27. What is This keyword ?**

It is used to point to the current object whenever the local variable and global variable names are same to differentiate between them we use “ this “ keyword.

## **28. What is Method / Function Overriding and Overloading in Java ?**

Developing multiple methods with the same name but variation in argument list is called as method overloading.

Developing a method in the sub class with the same name and signature as in the superclass but with different implementation in the sub class is called as method overriding.

**29. Define class?**

Class is a blueprint or a template to create object .

**30. What do you mean by Object?**

Object is real time entity which has its own state and behaviour.

**31. List the three steps for creating an Object for a class?**

An Object is first declared, then instantiated and then it is initialized.

**32. What is Singleton class?**

Singleton class control object creation, limiting the number to one but allowing the flexibility to create more objects if the situation changes.

**33. What is Inheritance ?**

Inheriting the property from one class to another class is called as Inheritance.

**34. Types of Inheritance ?**

- |                             |                              |
|-----------------------------|------------------------------|
| 1. single level Inheritance | 2. multi - level Inheritance |
| 3. hierarchical Inheritance | 4. multiple Inheritance      |
| 5. hybrid Inheritance       |                              |

**35. Is there any limitation of using Inheritance?**

Yes, since inheritance inherits everything from the super class and interface, it may make the subclass too clustering and sometimes error-prone when dynamic overriding or dynamic overloading in some situation.

**36. When super keyword is used?**

In the case of method overriding , along with the sub class implementation , if we need super class implementation thus we should go for super . method name

**37. Does Java support multiple inheritance ?**

No, Java does not support multiple inheritance. Each class is able to extend only on one class ,but is able to implement more than one interfaces.

**38. What is type casting?**

Converting from one type to another type is called Type casting.

**39. What is Polymorphism?**

An object showing different behaviour at different stages of its lifecycle is called polymorphism.

**40. What is compile time polymorphism ?**

The method declaration getting binded to its definition at the compile time by the compiler based on the arguments passed is called “ compile time polymorphism “.

**41. What is runtime polymorphism or dynamic method dispatch?**

The method declaration gets binded to its definition of its definition at the run time by the JVM based on the object created is called as “ run time polymorphism“.

**42. What is Dynamic Binding and late binding?**

Since the method declaration gets binded to its definition , at the rebinded , hence it is called as “ Dynamic binding “.

Since the method declaration gets binded to its definition , at the run time , hence it is called as “ late binding “.

**43. What is Abstraction?**

Hiding the complexity of the system and exposing only the required functionality to the end user is called as abstraction.

**44. What is Abstract class?**

Any class which is declared with a keyword abstract is called as “ Abstract class“.

**45. When Abstract methods are used?**

Any method which is declared with a keyword abstract is called as “Abstract method”.

**46. What is Encapsulation?**

It is one of the oops principles. Java is by default encapsulated.

**47. What is the primary benefit of Encapsulation?**

The main benefit of encapsulation is the ability to modify our implemented code without breaking the code of others who use our code. With this Encapsulation gives maintainability, flexibility and extensibility to our code.

**48. What is an Interface?**

Interface is by default abstract and it is a pure abstract body.

**49. Give some features of Interface?**

It includes – Interface cannot be instantiated An interface does not contain any constructors. All of the methods in an interface are abstract.

**50. Define Packages in Java?**

It is a folder structure which is used to store similar kind of files. The packages should be created always in the reverse order.

**51. Why Packages are used?**

Packages are used in Java in-order to prevent naming conflicts, to control access, to make searching/locating and usage of classes, interfaces, enumerations and annotations, etc., easier.

**52. What do you mean by Access Modifier?**

Access specifier is used to restrict the access from one class to another class ( or ) from one package to another package.

**53. What is Private access modifier?**

Any member of the class declared with a keyword private is called as private access specifier.

**54. What is Default / Package Level access modifier?**

If we declare any member of the class without any access specifier ( or ) keyword is called as default access specifier.

**55. What is protected access modifier?**

Any members of the class declared with the keyword protected is called as “protected member of the class “.

**56. What is public access modifier?**

Any members of the class declared with a keyword public is called as “public access specifier”.

**57. Why is String class considered immutable?**

When multiple reference variable are pointing to a single object and if one of the reference variable de reference with the current object it will not affect other reference variable. This is why it is immutable.

**58. Why is String Buffer called mutable?**

The String class is considered as immutable, so that once it is created a String object cannot be changed. If there is a necessity to make a lot of modifications to Strings of characters then String Buffer should be used.

**59. What is the difference between String Buffer and StringBuilder class?**

Use StringBuilder whenever possible because it is faster than String Buffer. But, if thread safety is necessary then use String Buffer objects.

**60. what is Thread ?**

Thread is an extension instance which owns of its own CPU time and memory.

**61. What do you mean by Multithreaded program?**

Processing multiple threads simultaneously is called as multi - threading.

**62. What is Sleep method ?**



Sleep method is a static method which pause its execution for few milliseconds.

**63. What is multi – tasking ?**

It is a Process of executing multiple task simultaneously is called as multi - tasking.

**64. What is Synchronized ?**

Processing one by one thread is called as Synchronized .

Processing multiple threads simultaneously is called as “non – Synchronized “.

**65. What are the two ways in which Thread can be created?**

Thread can be created by - implementing Runnable interface, extending the Thread class.

**66. What invokes a thread's run method?**

After a thread is started, via its start method of the Thread class, the JVM invokes the thread's run method when the thread is initially executed.

**67. Describe life cycle of thread?**

A thread is a execution in a program. The life cycle of a thread include – Newborn state, Runnable state, Running state, Blocked state, Dead state

**68. What is the difference between processes and threads ?**

A process is an execution of a program, while a Thread is a single execution sequence within a process. A process can contain multiple threads. A Thread is sometimes called a lightweight process.

**69. What is the difference between yielding and sleeping?**

When a task invokes its yield method, it returns to the ready state. When a task invokes its sleep method, it returns to the waiting state.

**70. How does multi-threading take place on a computer with a single CPU?**

The operating system's task scheduler allocates execution time to multiple tasks. By quickly switching between executing tasks, it creates the impression that tasks execute sequentially.

### **71.What is an Exception?**

It is an event triggered during the execution of the program which interrupt the execution and stops the execution abruptly / suddenly handling this event by using try and catch or throws is called as “Exception handling”.

### **72.What do you mean by Checked Exceptions?**

Any exception which is caught by the compiler at the compile time , this is called COMPILE TIME EXECEPTION.

### **73. Explain Runtime Exceptions?**

Any exception which is not caught at the compile time and found at the run time is called as RUN TIME EXCEPTION.

### **74. Which are the two subclasses under Exception class?**

The Exception class has two main subclasses : IO Exception class and Runtime Exception Class.

### **75. When throws keyword is used?**

It is used to throw the exception of throwable type. It will be written internally.

### **76. When throw keyword is used?**

An exception can be thrown, either a newly instantiated one or an exception that you just caught, by using throw keyword.

### **77. How finally used under Exception Handling?**

Finally block is block in exceptional handling which will get executed mandatory even though the exception is addressed or not addressed.

### **78. What is Null Pointer Exception?**

A Null Pointer Exception is thrown when calling the instance method of a null object, accessing or modifying the field of a null object etc.

**79. When Arithmetic Exception is thrown?**

The Arithmetic Exception is thrown when integer is divided by zero or taking the remainder of a number by zero. It is never thrown in floating-point operations.

**80. Which arithmetic operations can result in the throwing of an Arithmetic Exception?**

Integer / and % can result in the throwing of an Arithmetic Exception.

**81. If a variable is declared as private, where may the variable be accessed?**

A private variable may only be accessed within the class in which it is declared.

**82. What class of exceptions are generated by the Java run-time system?**

The Java runtime system generates Runtime Exception and Error exceptions.

**83. Can try statements be nested?**

Yes

**84. Is it necessary that each try block must be followed by a catch block?**

It is not necessary that each try block must be followed by a catch block. It should be followed by either a catch block or a finally block.

**85. What is the Collections API?**

Collections is an unified architecture which consist of interface and class.

**86. Does garbage collection guarantee that a program will not run out of memory?**

Garbage collection does not guarantee that a program will not run out of memory. It is possible for programs to use up memory resources faster than they are garbage collected. It is also possible for programs to create objects that are not subject to garbage collection.

**87. Explain Set Interface?**

It is a collection of element which cannot contain duplicate elements. The Set interface contains only methods inherited from Collection and adds the restriction that duplicate elements are prohibited.

**88. Explain Tree Set?**

It is a Set implemented when we want elements in a sorted order.

**89. What is Comparable Interface?**

It is used to sort collections and arrays of objects using the collections. sort and java.util. The objects of the class implementing the Comparable interface can be ordered.

**90. Why Vector class is used?**

The Vector class provides the capability to implement a growable array of objects. Vector proves to be very useful if you don't know the size of the array in advance, or you just need one that can change sizes over the lifetime of a program.

**91. What are the advantages of Array List over arrays?**

Array List can grow dynamically and provides more powerful insertion and search mechanisms than arrays.

**92. Why deletion in Linked List is fast than Array List?**

Deletion in linked list is fast because it involves only updating the next pointer in the node before the deleted node and updating the previous pointer in the node after the deleted node.

**93. How do you decide when to use Array List and Linked List?**

If you need to frequently add and remove elements from the middle of the list and only access the list elements sequentially, then LinkedList should be used. If you need to support random access, without inserting or removing elements from any place other than the end, then Array List should be used.

**94. What is a Values Collection View ?**

It is a collection returned by the values method of the Map Interface, It contains all the objects present as values in the map.

**95. Which package is used for pattern matching with regular expressions?**

java.util.regex package is used for this purpose.

**96. java.util.regex consists of which classes?**

java.util.regex consists of three classes – Pattern class, Matcher class and Pattern Syntax Exception class.

**97. What is finalize method?**

It is possible to define a method that will be called just before an object's final destruction by the garbage collector. This method is called finalize, and it can be used to ensure that an object terminates cleanly.

**98. What is the purpose of File class?**

It is used to create objects that provide access to the files and directories of a local file system.

**99. What is the difference between the Reader/Writer class hierarchy and the Input Stream/ Output Stream class hierarchy?**

The Reader/Writer class hierarchy is character-oriented, and the Input Stream / Output Stream class hierarchy is byte-oriented.

**100. What are Wrapper classes?**

These are classes that allow primitive types to be accessed as objects. Example: Integer, Character, Double, Boolean etc.

**101. What is JAR file?**

JAR files is Java Archive files and it aggregates many files into one. It holds Java classes in a library. JAR files are built on ZIP file format and have .jar file extension.

**102. What is a WAR file?**

This is Web Archive File and used to store XML, java classes, and Java Server pages. which is used to distribute a collection of Java Server Pages, Java Servlets, Java classes, XML files, static Web pages etc.

**103. What is the difference between object oriented programming language and object based programming language?**

Object based programming languages follow all the features of OOPs except Inheritance. JavaScript is an example of object based programming languages.

**104. Can a constructor be made final?**

No, this is not possible.

**105. What is final class?**

Final classes are created so the methods implemented by that class cannot be overridden. It can't be inherited.

**106. How many bits are used to represent Unicode, ASCII, UTF-16, and UTF-8 characters?**

Unicode requires 16 bits and ASCII require 7 bits. Although the ASCII character set uses only 7 bits, it is usually represented as 8 bits. UTF-8 represents characters using 8, 16, and 18 bit patterns. UTF-16 uses 16-bit and larger bit patterns.

**107. Which Java operator is right associative?**

The = operator is right associative.

**108. What is the difference between a break statement and a continue statement?**

A break statement results in the termination of the statement to which it applies switch ,for ,do ,or while. A continue statement is used to end the current loop iteration and return control to the loop statement.

**109. What are Class Loaders?**

A class loader is an object that is responsible for loading classes. The class Class Loader is an abstract class.

**110. What will happen if static modifier is removed from the signature of the main method?**

Program throws "No Such Method Error" error at runtime.

**111. What is the default value of an object reference declared as an instance variable?**

Null, unless it is defined explicitly.

**112. Can constructor be inherited?**

No, constructor cannot be inherited

**113. What is the difference between the >> and >>> operators?**

The >> operator carries the sign bit when shifting right. The >>> zero-fills bits that have been shifted out.

**114. Does Java allow Default Arguments?**

No, Java does not allow Default Arguments.

**115. Where import statement is used in a Java program?**

Import statement is allowed at the beginning of the program file after package statement.

**116. Can an Interface extend another Interface?**

Yes an Interface can inherit another Interface, for that matter an Interface can extend more than one Interface.

**117. Which object oriented Concept is achieved by using overloading and overriding?**

Polymorphism

**118. Can a double value be cast to a byte?**

Yes, a double value can be cast to a byte.

**119. Is there any need to import java. Lang package?**

No, there is no need to import this package. It is by default loaded internally by the JVM.

**120. What environment variables do I need to set on my machine in order to be able to run Java programs?**

CLASSPATH and PATH are the two variables.

**121. What is an enumeration?**

An enumeration is an interface containing methods for accessing the underlying data structure from which the enumeration is obtained. It allows sequential access to all the elements stored in the collection.

**122. What is difference between Path and Class path?**

Path and Class path are operating system level environment variables. Path is defines where the system can find the executables.exe files and class path is used to specify the location of .class files.

**123. What is an Applet ?**

A java applet is program that can be included in a HTML page and be executed in a java enabled client browser. Applets are used for creating dynamic and interactive web applications.

**124. Explain the life cycle of an Applet.**

An applet may undergo the following states:

- Init : An applet is initialized each time is loaded.
- Start : Begin the execution of an applet.
- Stop : Stop the execution of an applet.
- Destroy : Perform a final cleanup, before unloading the applet.

**125. What happens when an applet is loaded ?**

First of all, an instance of the applet's controlling class is created. Then, the applet initializes itself and finally, it starts running.



**126. The immediate superclass of the Applet class?**

Panel is the immediate superclass. A panel provides space in which an application can attach any other component, including other panels.

**127. Define canvas?**

It is a simple drawing surface which are used for painting images or to perform other graphical operations.

**128. Define Network Programming?**

It refers to writing programs that execute across multiple devices computers, in which the devices are all connected to each other using a network.

**129. What is a Socket?**

Sockets provide the communication mechanism between two computers using TCP. A client program creates a socket on its end of the communication and attempts to connect that socket to a server.

**130. Advantages of Java Sockets?**

Sockets are flexible and sufficient. Efficient socket based programming can be easily implemented for general communications. It cause low network traffic.

**131. Disadvantages of Java Sockets?**

Socket based communications allows only to send packets of raw data between applications. Both the client-side and server-side have to provide mechanisms to make the data useful in any way.

**132. Which class is used by server applications to obtain a port and listen for client requests?**

java.net.ServerSocket class is used by server applications to obtain a port and listen for client requests

**133. Which class represents the socket that both the client and server use to communicate with each other?**

java.net.Socket class represents the socket that both the client and server use to communicate with each other.

**134. What is the difference between a Choice and a List ?**

A Choice is displayed in a compact form that must be pulled down, in order for a user to be able to see the list of all available choices. Only one item may be selected from a Choice.

A List may be displayed in such a way that several List items are visible. A List supports the selection of one or more List items.

**135. What is a layout manager ?**

A layout manager is the used to organize the components in a container.

**136. What is the difference between a Scrollbar and a JScroll Pane ?**

A Scrollbar is a Component, but not a Container. A Scroll Pane is a Container. A Scroll Pane handles its own events and performs its own scrolling.

**137. Which Swing methods are thread-safe ?**

There are only three thread-safe methods: repaint, revalidate, and invalidate.

**138. Name three Component subclasses that support painting.**

The Canvas, Frame, Panel, and Applet classes support painting.

**139. What is clipping ?**

Clipping is defined as the process of confining paint operations to a limited area or shape.

**140. What is the difference between a Menu Item and a Check box Menu Item?**

The Check box Menu Item class extends the Menu Item class and supports a menu item that may be either checked or unchecked.

**141. How are the elements of a Border Layout organized ?**

The elements of a Border Layout are organized at the borders (North, South, East, and West) and the center of a container.

**142. How are the elements of a Grid Bag Layout organized ?**

The elements of a Grid Bag Layout are organized according to a grid. The elements are of different sizes and may occupy more than one row or column of the grid. Thus, the rows and columns may have different sizes.

**143. What is the difference between a Window and a Frame ?**

The Frame class extends the Window class and defines a main application window that can have a menu bar.

**144. What is the relationship between clipping and repainting ?**

When a window is repainted by the AWT painting thread, it sets the clipping regions to the area of the window that requires repainting.

**145. What is the relationship between an event-listener interface and an event adapter class ?**

An event-listener interface defines the methods that must be implemented by an event handler for a particular event. An event adapter provides a default implementation of an event-listener interface.

**146. How can a GUI component handle its own events ?**

A GUI component can handle its own events, by implementing the corresponding event-listener interface and adding itself as its own event listener.

**147. What advantage do Java's layout managers provide over traditional windowing systems ?**

Java uses layout managers to lay out components in a consistent manner, across all windowing platforms. Since layout managers aren't tied to absolute sizing and positioning, they are able to accommodate platform-specific differences among windowing systems.

**148. What is the design pattern that Java uses for all Swing components ?**

The design pattern used by Java for all Swing components is the Model View Controller (MVC) pattern.

**149. What is JDBC ?**

JDBC is an abstraction layer that allows users to choose between databases. JDBC enables developers to write database applications in Java, without having to concern themselves with the underlying details of a particular database.

**150. Explain the role of Driver in JDBC.**

The JDBC Driver provides vendor-specific implementations of the abstract classes provided by the JDBC API. Each driver must provide implementations for the following classes of the java. sql package : Connection, Statement, Prepared Statement, Callable Statement, Result Set and Driver.

**151. What is the purpose Class. for Name method ?**

This method is used to method is used to load the driver that will establish a connection to the database.

**152. What is the advantage of Prepared Statement over Statement ?**

Prepared Statements are precompiled and thus, their performance is much better. Also, Prepared Statement objects can be reused with different input values to their queries.

**153. What is the use of Call able Statement ? Name the method, which is used to prepare a Call able Statement.**

A Call able Statement is used to execute stored procedures. Stored procedures are stored and offered by a database. Stored procedures may take input values from the user and may return a result. The usage of stored procedures is highly encouraged, because it offers security and modularity. The method that prepares a Call able Statement is the following: Call able Statement. Prepare Call();

**154. What does Connection pooling mean ?**

The interaction with a database can be costly, regarding the opening and closing of database connections. Especially, when the number of database clients

increases, this cost is very high and a large number of resources is consumed. A pool of database connections is obtained at start up by the application server and is maintained in a pool. A request for a connection is served by a connection residing in the pool. In the end of the connection, the request is returned to the pool and can be used to satisfy future requests.

### **155. What is RMI ?**

The Java Remote Method Invocation (Java RMI) is a Java API that performs the object-oriented equivalent of remote procedure calls(RPC),with support for direct transfer of serialized Java classes and distributed garbage collection. Remote Method Invocation(RMI) can also be seen as the process of activating a method on a remotely running object. RMI offers location transparency because a user feels that a method is executed on a locally running object. Check some RMI Tips [here](#).

### **156. What is the basic principle of RMI architecture ?**

The RMI architecture is based on a very important principle which states that the definition of the behavior and the implementation of that behavior, are separate concepts. RMI allows the code that defines the behavior and the code that implements the behavior to remain separate and to run on separate JVMs.

### **157. What are the layers of RMI Architecture ?**

The RMI architecture consists of the following layers:

- **Stub and Skeleton layer:** This layer lies just beneath the view of the developer. This layer is responsible for intercepting method calls made by the client to the interface and redirect these calls to a remote RMI Service.
- **Remote Reference Layer:** The second layer of the RMI architecture deals with the interpretation of references made from the client to the server's remote objects. This layer interprets and manages references made from clients to the remote service objects. The connection is a one-to-one (unicast) link.
- **Transport layer:** This layer is responsible for connecting the two JVM participating in the service. This layer is based on TCP/IP connections between machines in a network. It provides basic connectivity, as well as some firewall penetration strategies.

**158. What is the role of Remote Interface in RMI ?**

The Remote interface serves to identify interfaces whose methods may be invoked from a non-local virtual machine. Any object that is a remote object must directly or indirectly implement this interface. A class that implements a remote interface should declare the remote interfaces being implemented, define the constructor for each remote object and provide an implementation for each remote method in all remote interfaces.

**159. What is the role of the java. rmi. Naming Class ?**

The java. rmi. Naming class provides methods for storing and obtaining references to remote objects in the remote object registry. Each method of the Naming class takes as one of its arguments a name that is a String in URL format.

**160 What is meant by binding in RMI ?**

Binding is the process of associating or registering a name for a remote object, which can be used at a later time, in order to look up that remote object. A remote object can be associated with a name using the bind or rebind methods of the Naming class.

**161. What is the difference between using bind() and rebind() methods of Naming Class ?**

The bind method bind is responsible for binding the specified name to a remote object, while the rebind method is responsible for rebinding the specified name to a new remote object. In case a binding exists for that name, the binding is replaced.

**162. What are the steps involved to make work a RMI program ?**

The following steps must be involved in order for a RMI program to work properly:

- Compilation of all source files.
- Generation of the stubs using rmic.
- Start the rmi registry.

- Start the RMI Server.
- Run the client program.

**163. What is the role of stub in RMI ?**

A stub for a remote object acts as a client's local representative or proxy for the remote object. The caller invokes a method on the local stub, which is responsible for executing the method on the remote object. When a stub's method is invoked, it undergoes the following steps:

- It initiates a connection to the remote JVM containing the remote object.
- It marshals the parameters to the remote JVM.
- It waits for the result of the method invocation and execution.
- It unmarshals the return value or an exception if the method has not been successfully executed.
- It returns the value to the caller.

**164. What is DGC ? And how does it work ?**

DGC stands for Distributed Garbage Collection. Remote Method Invocation (RMI) uses DGC for automatic garbage collection. Since RMI involves remote object references across JVM's, garbage collection can be quite difficult. DGC uses a reference counting algorithm to provide automatic memory management for remote objects.

**165. What is the purpose of using RMI Security Manager in RMI ?**

RMI Security Manager provides a security manager that can be used by RMI applications, which use own loaded code. The class loader of RMI will not download any classes from remote locations, if the security manager has not been set.

**166. Explain Marshalling and demarshalling.**

When an application wants to pass its memory objects across a network to another host or persist it to storage, the in-memory representation must be

converted to a suitable format. This process is called marshalling and the revert operation is called demarshalling.

### **167. Explain Serialization and Deserialization.**

Java provides a mechanism, called object serialization where an object can be represented as a sequence of bytes and includes the object's data, as well as information about the object's type, and the types of data stored in the object. Thus, serialization can be seen as a way of flattening objects, in order to be stored on disk, and later, read back and reconstituted. Deserialization is the reverse process of converting an object from its flattened state to a live object.

### **168. What is a Servlet ?**

The servlet is a Java programming language class used to process client requests and generate dynamic web content. Servlets are mostly used to process or store data submitted by an HTML form, provide dynamic content and manage state information that does not exist in the stateless HTTP protocol.

### **169. Explain the architecture of a Servlet.**

The core abstraction that must be implemented by all servlets is the `javax.servlet.Servlet` interface. Each servlet must implement it either directly or indirectly, either by extending `javax.servlet.GenericServlet` or `javax.servlet.http.HttpServlet`. Finally, each servlet is able to serve multiple requests in parallel using multithreading.

### **170. What is the difference between an Applet and a Servlet ?**

An Applet is a client side java program that runs within a Web browser on the client machine. On the other hand, a servlet is a server side component that runs on the web server. An applet can use the user interface classes, while a servlet does not have a user interface. Instead, a servlet waits for client's HTTP requests and generates a response in every request.

### **171. What is the difference between Generic Servlet and Http Servlet ?**

Generic Servlet is a generalized and protocol-independent servlet that implements the `Servlet` and `Servlet Config` interfaces. Those servlets extending the `Generic Servlet` class shall override the `service` method. Finally, in order to develop



an HTTP servlet for use on the Web that serves requests using the HTTP protocol, your servlet must extend the `Http Servlet` instead. Check Servlet examples [here](#).

### **172. Explain the life cycle of a Servlet.**

On every client's request, the Servlet Engine loads the servlets and invokes its init methods, in order for the servlet to be initialized. Then, the Servlet object handles all subsequent requests coming from that client, by invoking the service method for each request separately. Finally, the servlet is removed by calling the server's destroy method.

### **173. What is the difference between `doGet()` and `doPost()` ?**

**Do GET:** The GET method appends the name-value pairs on the request's URL. Thus, there is a limit on the number of characters and subsequently on the number of values that can be used in a client's request. Furthermore, the values of the request are made visible and thus, sensitive information must not be passed in that way.

**Do POST :** The POST method overcomes the limit imposed by the GET request, by sending the values of the request inside its body. Also, there is no limitations on the number of values to be sent across. Finally, the sensitive information passed through a POST request is not visible to an external client.

### **174. What is meant by a Web Application ?**

A Web application is a dynamic extension of a Web or application server. There are two types of web applications: presentation oriented and service-oriented. A presentation-oriented Web application generates interactive web pages, which contain various types of markup language and dynamic content in response to requests. On the other hand, a service-oriented web application implements the endpoint of a web service. In general, a Web application can be seen as a collection of servlets installed under a specific subset of the server's URL namespace.

### **175. What is a Server Side Include (SSI) ?**

Server Side Includes (SSI) is a simple interpreted server-side scripting language, used almost exclusively for the Web, and is embedded with a servlet tag. The most frequent use of SSI is to include the contents of one or more files into a

Web page on a Web server. When a Web page is accessed by a browser, the Web server replaces the servlet tag in that Web page with the hypertext generated by the corresponding servlet.

### **176. What is Servlet Chaining ?**

Servlet Chaining is the method where the output of one servlet is sent to a second servlet. The output of the second servlet can be sent to a third servlet, and so on. The last servlet in the chain is responsible for sending the response to the client.

### **177. How do you find out what client machine is making a request to your servlet ?**

The Servlet Request class has functions for finding out the IP address or host name of the client machine. Get Remote Addr() gets the IP address of the client machine and get Remote Host() gets the host name of the client machine.

### **178. What is the structure of the HTTP response ?**

The HTTP response consists of three parts:

- **Status Code:** describes the status of the response. It can be used to check if the request has been successfully completed. In case the request failed, the status code can be used to find out the reason behind the failure. If your servlet does not return a status code, the success status code, Http Servlet Response. SC\_OK, is returned by default.
- **HTTP Headers:** they contain more information about the response. For example, the headers may specify the date/time after which the response is considered state, or the form of encoding used to safely transfer the entity to the user. See how to retrieve headers in Servlet [here](#).
- **Body:** it contains the content of the response. The body may contain HTML code, an image, etc. The body consists of the data bytes transmitted in an HTTP transaction message immediately following the headers.

### **179. What is a cookie ? What is the difference between session and cookie ?**

A cookie is a bit of information that the Web server sends to the browser. The browser stores the cookies for each Web server in a local file. In a future request, the browser, along with the request, sends all stored cookies for that specific Web server. The differences between session and a cookie are the following:

- The session should work, regardless of the settings on the client browser. The client may have chosen to disable cookies. However, the sessions still work, as the client has no ability to disable them in the server side.
- The session and cookies also differ in the amount of information they can store. The HTTP session is capable of storing any Java object, while a cookie can only store String objects.

### **180. Which protocol will be used by browser and servlet to communicate ?**

The browser communicates with a servlet by using the HTTP protocol.

### **181. What is HTTP Tunneling ?**

HTTP Tunneling is a technique by which, communications performed using various network protocols are encapsulated using the HTTP or HTTPS protocols. The HTTP protocol therefore acts as a wrapper for a channel that the network protocol being tunneled uses to communicate. The masking of other protocol requests as HTTP requests is HTTP Tunneling.

### **182. What's the difference between send Redirect and forward methods ?**

The send Redirect method creates a new request, while the forward method just forwards a request to a new target. The previous request scope objects are not available after a redirect, because it results in a new request. On the other hand, the previous request scope objects are available after forwarding. Finally, in general, the send Redirect method is considered to be slower compared to the forward method.

### **182. What is URL Encoding and URL Decoding ?**

The URL encoding procedure is responsible for replacing all the spaces and every other extra special character of a URL, into their corresponding Hex representation. In correspondence, URL decoding is the exact opposite procedure.

**183. What is a JSP Page ?**

A Java Server Page (JSP) is a text document that contains two types of text: static data and JSP elements. Static data can be expressed in any text-based format, such as HTML or XML. JSP is a technology that mixes static content with dynamically generated content. See JSP example [here](#).

**184. How are the JSP requests handled ?**

On the arrival of a JSP request, the browser first requests a page with a .jsp extension. Then, the Web server reads the request and using the JSP compiler, the Web server converts the JSP page into a servlet class. Notice that the JSP file is compiled only on the first request of the page, or if the JSP file has changed. The generated servlet class is invoked, in order to handle the browser's request. Once the execution of the request is over, the servlet sends a response back to the client. See how to get Request parameters in a JSP.

**185. What are the advantages of JSP ?**

The advantages of using the JSP technology are shown below:

- JSP pages are dynamically compiled into servlets and thus, the developers can easily make updates to presentation code.
- JSP pages can be pre-compiled.
- JSP pages can be easily combined to static templates, including HTML or XML fragments, with code that generates dynamic content.
- Developers can offer customized JSP tag libraries that page authors access using an XML-like syntax.
- Developers can make logic changes at the component level, without editing the individual pages that use the application's logic.

**186. What are Directives ? What are the different types of Directives available in JSP ?**

Directives are instructions that are processed by the JSP engine, when the page is compiled to a servlet. Directives are used to set page-level instructions, insert data from external files, and specify custom tag libraries. Directives are

defined between `< %@` and `% >`. The different types of directives are shown below:

- Include directive: it is used to include a file and merges the content of the file with the current page.
- Page directive: it is used to define specific attributes in the JSP page, like error page and buffer.
- Tag lib: it is used to declare a custom tag library which is used in the page.

### **187. What are JSP actions ?**

JSP actions use constructs in XML syntax to control the behavior of the servlet engine. JSP actions are executed when a JSP page is requested. They can be dynamically inserted into a file, re-use JavaBeans components, forward the user to another page, or generate HTML for the Java plugin. Some of the available actions are listed below:

- `jsp : include` - includes a file, when the JSP page is requested.
- `jsp : useBean` - finds or instantiates a JavaBean.
- `jsp : setProperty` - sets the property of a JavaBean.
- `jsp : getProperty` - gets the property of a JavaBean.
- `jsp : forward` - forwards the requester to a new page.
- `jsp : plugin` - generates browser-specific code.

### **188. What are Scriptlets ?**

In Java Server Pages (JSP) technology, a scriptlet is a piece of Java-code embedded in a JSP page. The scriptlet is everything inside the tags. Between these tags, a user can add any valid scriptlet.

### **189. What are Declarations ?**

Declarations are similar to variable declarations in Java. Declarations are used to declare variables for subsequent use in expressions or scriptlets. To add a declaration, you must use the sequences to enclose your declarations.

**190. What are Expressions ?**

A JSP expression is used to insert the value of a scripting language expression, converted into a string, into the data stream returned to the client, by the web server. Expressions are defined between `<% =` and `%>` tags.

**191. What is meant by implicit objects and what are they ?**

JSP implicit objects are those Java objects that the JSP Container makes available to developers in each page. A developer can call them directly, without being explicitly declared. JSP Implicit Objects are also called pre-defined variables. The following objects are considered implicit in a JSP page:

- application
- page
- request
- response

**192. Which package has light weight components?**

javax.Swing package. All components in Swing, except JApplet, JDialog, JFrame and JWindow are lightweight components.